IntelliVault Model Training Notes

1. Overview

IntelliVault is a secure document intelligence system that uses a Retrieval-Augmented Generation (RAG) pipeline to provide accurate and context-aware responses. The system combines document storage, embeddings, vector search, and large language models.

2. Data Collection

- Gather high-quality, domain-specific documents.

- Ensure data is clean, structured, and relevant.

- Remove duplicates and sensitive/unnecessary information.

3. Data Preprocessing

- Convert documents into text format.

- Chunk documents into smaller segments.

- Clean text (remove noise, special characters, etc.).

4. Embedding Generation

- Use embedding models (e.g., sentence transformers).

- Convert text chunks into vector representations.

- Store embeddings in a vector database.

5. Vector Database

- Use tools like FAISS, Pinecone, or ChromaDB.

- Store embeddings for efficient similarity search.

- Enable fast retrieval of relevant documents.

6. Retrieval Process

- Convert user query into embedding.
- Perform similarity search in vector DB.
- Retrieve top-k relevant chunks.

7. Prompt Engineering

- Combine retrieved context with user query.
- Design prompts to guide the LLM effectively.
- Ensure answers are grounded in retrieved data.

8. Model Selection

- Use LLMs like GPT, LLaMA, or HuggingFace models.
- Choose based on performance, cost, and latency.

9. Fine-Tuning (Optional)

- Train model on domain-specific data.
- Improve accuracy and relevance.

10. Evaluation

- Measure accuracy, relevance, and response time.
- Use test queries and benchmark datasets.

11. Security Considerations

- Encrypt sensitive data.
- Implement access control.

- Ensure secure API usage.

12. Deployment

- Use cloud platforms (AWS, Azure, GCP).

- Monitor performance and logs.

- Scale as needed.

13. Continuous Improvement

- Collect user feedback.

- Update dataset regularly.

- Improve prompts and retrieval strategies.

Conclusion

Training IntelliVault involves integrating data processing, embeddings, retrieval, and LLMs into a cohesive pipeline. Continuous optimization ensures better performance and security.